Working Glider Notes

Glider Operating States

Pico (persistor based glider)/Shell (stm32 based glider)

- The base operating system
- **DO NOT USE NEAR ANY BODY OF WATER**

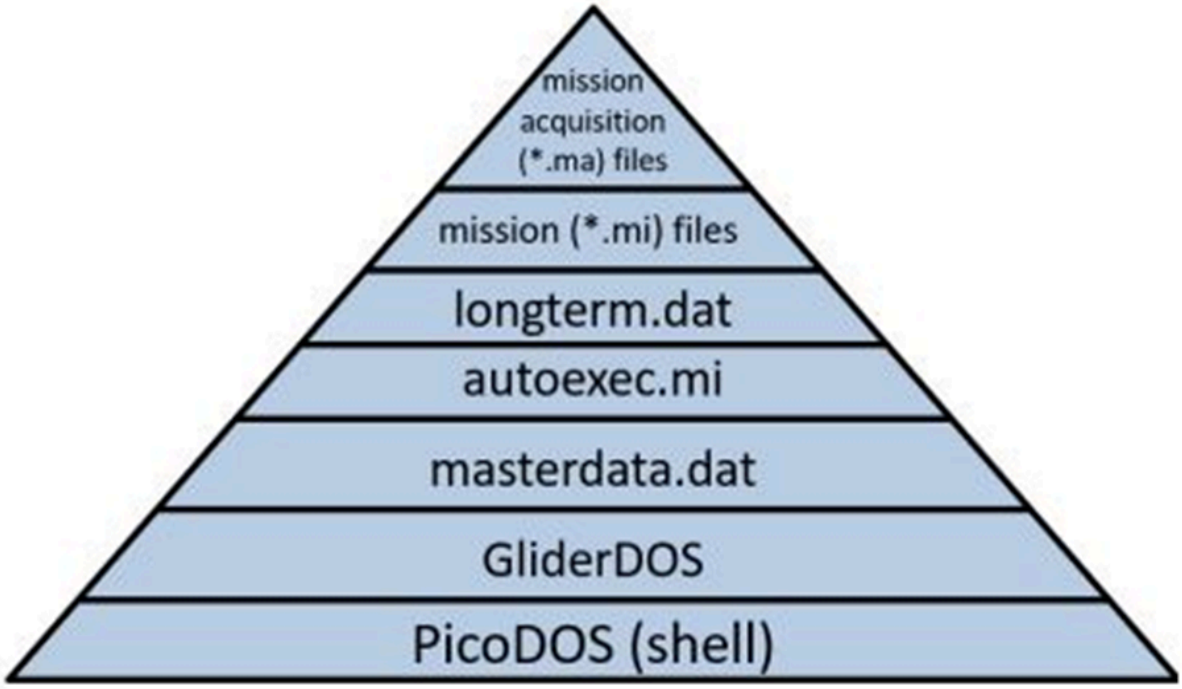
GliderDos

- The glider application that is run by processors.
- The glider can use important devices in this state.

GliderLab

- The glider application that is run by processors.
- The glider can use important devices in this state.
- **SAME AS GliderDos BUT SOME SAFETY FEATURES ARE DISABLED.**
- **DO NOT USE IN WATER.**

Glider Software Hierarchy



Glider Software Hierarchy Cont.

Masterdata.dat

- Master repository for all default of the variables.
- Stores all the default values that the glider uses.
- **DO NOT CHANGE THIS FILE.**

Autoexec.mi

- The glider identification file.
- Overwrites a subset of variables from the Masterdata.dat file.
    - Some variables will change over time
- Stores glider calibration numbers
- Add comments to the top of the Autoexec.mi file by using #.
- The name of the glider can be changed here by altering the name variable.
- Lists the installed hardware.
- Lists the preset settings.
- Numbers might need to be changed after a calibration is performed on the hardware.
- You can find the iridium phone numbers here.
- **NEVER CHANGE THIS FILE OVER IRIDIUM.**

Longterm.data

- Holds variables that change from power cycle to power cycle.
- **Keep track of m_tot_num_inflections.**
    - This is needed to determine when your ballast pumps should be serviced.

Mission(*.mi) files

- Files that dictate how the glider will operate during a mission.
- astock.mi is the default mission file from Teledyne.
    - Default mission
    - Mission is for glider to fly around 4 points in Ashument Pond.
        - Where Teledyne Webb Research regularly flies to test gliders.
    - **Always keep a version of the default file.**
- Calls **behavior_files.ma**

Glider Software Hierarchy Cont.

Behavior: Surface Behaviors

File Name	Description
Surfac21.ma	**Surface if there are no communications for a certain amount of time.**
Surfac22.ma	**Surface if the glider has no more waypoints to go to.**
Surfac23.ma	**Surface when yo's get caught in a loop. Surface when no further pitch command is given.**
Surfac24.ma	**Surface when glider thinks that it hit a waypoint through dead reckoning.**
Surfac25.ma	**Science wants to surface.**
Surfac26.ma	**Surface after a certain amount of time.**

Waypoint and yo's

File Name	Description
Goto_l10.ma	**Waypoint management file.**
Yo14.ma	**Number of half yo's for the glider to do before surfacing.**

Behavior Arguments

Abend (Abort) Behavior Arguments

Argument Name (b_arg)	Description	Arguments
overdepth(m)	The max working depth of the glider. Value is clipped to: F_MAX_WORKING_DEPTH	Set number in meters. **Example: 1000** **<0 will disable over depth.**
overdepth_sample_times(s)	How often the glider will check to see if it's over depth.	Number of seconds between each check. **Example: 10.0**
overtime(s)	The bool to abort if overtime.	**-1** enables abort. **<0** disables.
samedepth_for(s)	Abort if glider is at the same depth for a certain amount of time.	In seconds. **Example: 120.0** **<0 will disable.**
samedepth_for_sample_times(s)	How often the glider will check to see if the glider is at the same depth for a certain amount of time.	In seconds. **Example: 30.0**
no_cop_tickle_for(s)	Abort mission if watchdog is not tickled after a certain time. Is the glider talking to the computer.	In seconds. **Example: 7000.0** **<0 will disable.**

Surface Behavior Arguments

Argument Name (b_arg)	Description	Arguments
args_from_file(enum)	Reads from mafiles/surfac#.ma	Surfac.ma file number **Example: 10**

Configuration Files

Files in the Config Folder

File Name	Description
appcmd.dat	Simulates functions in the payload bay. The file must be deleted before an actual flight.